Data Archiving Strategy — Microsoft Purview

Implementation Solution Guide

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Executive Summary

The problem

Cold, infrequently accessed data clogs primary mail and content stores. Left unaddressed, it drives up storage cost, slows the environment, and becomes a **compliance gap and data-risk exposure** — data nobody manages, on the most expensive tier, with no defined lifecycle.

The solution

This solution delivers an end-to-end **data archiving strategy** in Microsoft Purview and Exchange Online. It moves cold data off primary stores **without losing it and without breaking access**, using three coordinated levers:

- **Mailbox archiving** — Exchange Online archive mailboxes (and auto-expanding archive where volume warrants), keeping cold mail fully searchable in a lower-cost tier.
- **Retention / lifecycle policy** — Messaging Records Management move-to-archive tags that age content into the archive automatically, plus Purview Data Lifecycle Management retention policies that keep (and, where the records schedule requires, dispose of) content on a defined timeline.
- **Tiering** — a hot → warm → cold model so storage cost tracks how often data is actually used.

How it is delivered

Every high-impact change is **piloted on a limited, named scope before tenant-wide expansion**: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**. The pilot uses test mailboxes, disables all delete actions, and gates every write behind an explicit confirmation, because archiving moves data and is not trivially reversible. Configuration is proven with **pass/fail gates** and captured **evidence** at each step.

What is included

Seven guides (Overview, Requirements, Implementation, Testing, Troubleshooting, Best Practices, Deliverables) and **five PowerShell tools** (Discovery, Configuration/Remediation, Validation, Reporting, and a scope/health helper) — all read-only or preview-mode by default, with no credentials in code.

Boundaries and assurances

- **Records / compliance / legal own the retention timeline** and any disposition decision; this solution provides the technical method and the evidence, not the determination.
- Microsoft Purview and Exchange archiving features are **licensing-/plan-dependent and evolving** — availability and paths are verified against current Microsoft documentation, never invented.
- Automation is **PowerShell** with modern-auth sign-in and **no hardcoded secrets**.

The bottom line

Cold data is archived — **contained, still accessible, cost-optimized, validated, and defensible** — in Microsoft Purview.

Solution Overview

What this delivers

This solution delivers a **data archiving strategy** in Microsoft Purview and Exchange Online: a defensible way to move **cold, infrequently accessed data** off primary stores into lower-cost archive tiers **without losing it and without breaking access**. It combines three levers:

1. **Mailbox archiving** — Exchange Online **archive mailboxes** (In-Place Archive) and, where

volume warrants, **auto-expanding archive**, so a user's cold mail moves out of the primary mailbox into archive storage that is still fully searchable.

1. **Retention / lifecycle policy** — **Messaging Records Management (MRM)** *move-to-archive* tags

and policies that age content into the archive automatically, plus **Microsoft Purview Data Lifecycle Management** retention policies that keep (and, where the records schedule requires, dispose of) content on a defined timeline.

1. **Tiering** — a hot → warm → cold model (primary mailbox → archive mailbox → cold/archive

storage such as **Microsoft 365 Archive** for SharePoint), so the storage cost of data tracks its access frequency.

The outcome is a configured, **piloted**, and **validated** archiving capability with named policies, pass/fail gates, evidence, and a documented rollback — so cold data is contained and the strategy is operable and defensible.

> ***Scope boundary.*** *This solution provides the* ***technical configuration method and evidence****. The* ***retention schedule itself*** *— how long each class of content must be kept, when it may be disposed, and what is a business record — is a* ***records / compliance / legal determination*** *and is* ***deferred*** *to those functions (see the category Global Requirements). We configure what they decide; we do not decide it.*

How it works in Microsoft Purview and Exchange Online

• **Archive mailbox (In-Place Archive)** is an additional mailbox that appears alongside the primary

mailbox in Outlook / Outlook on the web. Enabling it requires **Exchange Online Plan 2** (included in Microsoft 365 E3/E5) or **Exchange Online Plan 1 with the Exchange Online Archiving add-on** — *verify your licensing/tenant.*

• **Auto-expanding archive** adds storage automatically as the archive fills, for users with very

large archives. It is enabled per-mailbox (`Enable-Mailbox -AutoExpandingArchive`) or org-wide (`Set-OrganizationConfig -AutoExpandingArchive $true`). **Enabling it is not reversible** — pilot deliberately.

• **MRM move-to-archive tags** (retention tags with `RetentionAction MoveToArchive`) are linked into

a **retention policy** assigned to mailboxes; the **Managed Folder Assistant (MFA)** processes mailboxes and moves aged items into the archive.

• **Purview Data Lifecycle Management** retention policies and labels apply *keep* / *keep-and-delete* / *delete* actions across Exchange, SharePoint, OneDrive and Teams from the Purview portal.

• **Tiering to cold storage** — **Microsoft 365 Archive** moves inactive SharePoint sites to a cold storage tier (pay-as-you-go via an Azure subscription). It is **evolving / billing-gated** — treat paths and availability as *verify your licensing/tenant.*

The delivery model

This solution follows the standard Purview delivery model, and **every high-impact change is piloted on a limited scope before tenant-wide expansion**:

Phase	What happens
Assess	Inventory mailbox sizes, existing archive status, current MRM tags/policies, existing DLM retention policies, and the org-level auto-expanding archive setting. Confirm licensing and roles.
Design	Define the tiering model (what ages to archive, at what age), name the MRM tags and policy, and capture the retention timeline **as decided by records/compliance** — not by us.
Configure	Create the MRM move-to-archive tag and policy; enable archive mailboxes for the pilot scope; stage the DLM retention policy against the pilot scope. Scripts default to **read-only /** `-WhatIf`.
Pilot	Apply to a **named pilot group** of test mailboxes only. Run the Managed Folder Assistant and observe items moving to archive. **No delete action** is enabled in pilot.
Validate	Run pass/fail checks: archive enabled, policy assigned, MFA processed, items in archive, retention policy applied. Capture evidence.
Expand (enforce)	Widen scope to production mailboxes under change control once the pilot passes.

> ***Reversibility caveat.*** *Archiving moves data. Once items are moved into the archive or an auto-expanding archive is enabled, the change is* ***not trivially reversible****, and any *delete* action is irreversible. This is why the pilot uses test mailboxes, disables delete actions, and gates all write operations behind an explicit* `-Commit` *switch.*

What is included

• **Solution Overview** (this document).

• **Requirements Guide** — solution-specific prerequisites with verify/implement steps.

• **Implementation Guide** — exact ordered configuration steps, every policy and tag named, with pass/fail gates and rollback.

• **Testing & Validation** — the gates that prove the archiving strategy works.

• **Troubleshooting** — common failure signatures and resolutions.

• **Best Practices** — how to keep the strategy healthy and defensible.

• **5 PowerShell scripts** — Discovery, Configuration/Remediation, Validation, Reporting, and a scope/health helper (Exchange Online + Security & Compliance PowerShell; read-only or `-WhatIf` by default; no credentials in code).

Requirements

These are the **solution-specific** prerequisites for the data archiving strategy. They are **in addition to** the six category-wide prerequisites in the **Purview Global Requirements** (`global-requirements-purview.md`) — tenant licensing, least-privilege roles, the S&C PowerShell + Graph toolchain, in-scope data sources, compliance/records/legal stakeholders, and a pilot scope + change control. Verify each item below before starting the implementation steps.

> *Every capability here is **licensing-/plan-dependent** and the portals **evolve**. Where a plan or path is called out, **verify your licensing/tenant** rather than assuming availability.*

R1. Exchange Online archiving licensing (archive mailbox + auto-expanding archive)

• **Description:** Each user who will get an archive mailbox needs a plan that includes archiving:

Exchange Online Plan 2 (included in **Microsoft 365 E3 and E5**) or **Exchange Online Plan 1 plus the Exchange Online Archiving add-on**. **Auto-expanding archive** requires the same Plan 2 / add-on entitlement.

• **Why required:** Without the entitlement, `Enable-Mailbox -Archive` will not provision an archive, and auto-expanding archive is unavailable.

• **How to verify:** Microsoft 365 admin center → **Billing** → **Licenses** confirm E3/E5 or the archiving add-on; in Exchange Online PowerShell `Get-Mailbox <user> | fl ArchiveStatus,ArchiveGuid` and `Get-OrganizationConfig | fl AutoExpandingArchive`.

• **How to implement if missing:** Assign the correct licensing before enabling archives. Confirm the archive feature is visible in the Purview / Exchange admin surfaces — *verify your licensing/tenant.*

R2. Least-privilege roles for archiving and lifecycle

• **Description:** The engineer needs the specific roles for the two surfaces:

• **Exchange Online** — a role group with **Mail Recipients** and **Recipient Policies** management

(e.g. **Recipient Management**) to enable archives and manage MRM tags/policies, and permission to run the **Managed Folder Assistant**.

• **Microsoft Purview** — **Compliance Administrator** or **Compliance Data Administrator** (or the scoped **Retention Management** role) to create and publish **Data Lifecycle Management** retention policies/labels.

• **Global Administrator** only where genuinely required (e.g. org-wide `Set-OrganizationConfig`).

• **Why required:** Purview and Exchange use fine-grained role groups; the matching role is required for each configuration action.

• **How to verify:** Purview portal → **Roles & scopes** → **Permissions**; Exchange admin center → **Roles** → **Admin roles**. Confirm membership.

• **How to implement if missing:** Assign the least-privilege role group under change control; avoid standing Global Administrator.

R3. Managed Folder Assistant / MRM understanding and toolchain

• **Description:** A workstation with the **ExchangeOnlineManagement** module (for `Connect-ExchangeOnline` **and** `Connect-IPPSSession`) and **Microsoft Graph PowerShell**, signed in with a **modern-auth / MFA** admin account. The engineer must understand that MRM archiving is driven by the **Managed Folder Assistant (MFA)**, which runs on its own work cycle and can be triggered on demand with `Start-ManagedFolderAssistant`.

• **Why required:** Configuration, validation, and the provided scripts run through these modules; archiving does not take effect until the MFA processes the mailbox.

• **How to verify:** `Connect-ExchangeOnline`, `Connect-IPPSSession`, and `Connect-MgGraph` all succeed; `Get-Command Start-ManagedFolderAssistant` resolves.

• **How to implement if missing:** `Install-Module ExchangeOnlineManagement` and `Install-Module Microsoft.Graph`; confirm modern-auth sign-in.

R4. In-scope mailboxes / sites and the tiering intent

• **Description:** The **in-scope workloads** (which mailboxes get archives; whether SharePoint cold tiering is in scope) and a **tiering intent**: at what **age** cold content should move to archive (e.g. items older than **2 years** move to archive), and what stays in the primary store.

• **Why required:** MRM move-to-archive tags key off an **age limit**, and the archive scope keys off the mailbox set; both must be defined before authoring policy.

• **How to verify:** Confirm the pilot mailbox list, the target age threshold, and whether SharePoint **Microsoft 365 Archive** tiering is in scope with stakeholders.

• **How to implement if missing:** Define the scope and the age threshold before creating tags.

R5. Records / compliance / legal sign-off on the retention timeline

• **Description:** Named approval from **records management / compliance / legal** for the **retention timeline **and for** any disposition (delete) action **— including confirmation of any** legal holds** that must suppress deletion.

• **Why required:** The archiving *mechanism* is technical; **how long to keep and when to dispose is a records/legal determination** and must not be set to a technical default. Deletion is irreversible and can conflict with legal hold.

• **How to verify:** Confirm the approved retention schedule (or that only *keep/move-to-archive* is in scope for now), and confirm the legal-hold position with legal.

• **How to implement if missing:** Do **not** enable any delete action. Configure archiving (move + keep) only, and defer disposition until records/legal has approved a schedule.

R6. A named pilot scope, change window, and rollback plan

• **Description:** A **named pilot group** of test mailboxes (never production users for the first ring), an agreed **change window**, and a **rollback plan** that accounts for the fact that archiving is **not trivially reversible**.

• **Why required:** Every high-impact archiving change is piloted first; enabling auto-expanding archive and moving items are hard to undo, and delete is irreversible.

• **How to verify:** Confirm the pilot group membership, the approval path, and that the rollback plan is documented (see the Implementation Guide's rollback section).

• **How to implement if missing:** Define the pilot group and change control, and disable all delete actions, before touching production.

Global Requirements — Microsoft Purview Solutions

These requirements apply to **every** solution in the **Purview** category (implementing and managing Microsoft Purview, sensitivity & retention labels, DLP, data lifecycle & records, insider risk, communication compliance, audit, data security posture management, and data governance). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Purview solution).*** *The **Microsoft Purview portal** (purview.microsoft.com), the **Microsoft 365 admin center**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Microsoft Purview features are heavily **licensing-/plan-dependent** (many require **Microsoft 365 E5** or the **E5 Compliance** add-on; some are in preview) and the portal is **actively evolving** — capability availability and portal paths are **cited or flagged "verify your licensing/tenant"**, never asserted or invented. **Regulatory, records-retention, and legal determinations** (retention schedules, records file plans, legal holds, insider-risk case actions, regulatory applicability) are **deferred to the organization's compliance, records, privacy,***

> ***and legal functions*** *— the solutions provide the technical configuration method and evidence, not the determination. Automation is* ***PowerShell*** *(Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with the required Purview/compliance licensing

- **Description:** A Microsoft 365 tenant with the **compliance licensing** the capability requires

(E3 for baseline; **E5 / E5 Compliance** for advanced DLP, auto-labeling, insider risk, communication compliance, Audit Premium, DSPM, etc.).

- **Why required:** Most Purview capabilities are plan-gated; the feature won't be available without the

right plan.

- **How to verify:** In the M365 admin center confirm the subscription/add-ons; check the feature is

visible in the Purview portal.

- **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Purview roles (least-privilege) and portal access

- **Description:** Access with the least-privilege **Purview / compliance role** the solution names

(e.g. Compliance Administrator, Compliance Data Administrator, Information Protection, Records Management, Insider Risk, eDiscovery, Audit roles), with Global Administrator only where required.

- **Why required:** Purview uses fine-grained role groups; configuration requires the matching role.
- **How to verify:** Purview portal → **Roles & scopes / Permissions** — confirm the role membership.
- **How to implement if missing:** Assign the least-privilege role group under change control.

3. Security & Compliance PowerShell + Graph toolchain

- **Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the

ExchangeOnlineManagement module) and **Microsoft Graph PowerShell**, modern-auth / MFA-capable admin.

- **Why required:** Configuration, automation, and the provided scripts run through these.
- **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` succeed with the admin account.
- **How to implement if missing:** Install the modules and confirm modern-auth sign-in before running steps.

4. Data sources & scope in scope

- **Description:** The **workloads and data sources** in scope (Exchange, SharePoint, OneDrive, Teams,

endpoints/devices, and — for data governance — the broader data estate) and a **data-classification / sensitivity** intent to drive labels, DLP, and retention.

- **Why required:** Purview policies key off the in-scope workloads and the classification scheme.

• **How to verify:** Confirm the in-scope workloads and the classification/sensitivity approach with stakeholders.

• **How to implement if missing:** Define the scope and classification intent before authoring policies.

5. Compliance, records, privacy & legal stakeholders

• **Description:** Engagement with the **compliance, records-management, privacy, security, and legal**

stakeholders whose policies the configuration must satisfy (retention schedules, records file plans, regulatory obligations, insider-risk and communication-compliance policy, legal hold).

• **Why required:** Purview configuration must align to enterprise policy and legal/records requirements, not technical defaults; several capabilities have HR/legal sensitivity.

• **How to verify:** Confirm the applicable policies, the records schedule, and named approvers.

• **How to implement if missing:** Identify the owners and capture the requirements before configuring.

6. A pilot scope and change control

• **Description:** A **pilot scope** (users/sites/policies in simulation or a limited ring) to validate on before tenant-wide enforcement, and a **change-control** process — most Purview policies support a **simulation / test / disabled** mode that MUST be used before enforcement.

• **Why required:** Every solution follows an **Assess → Design → Configure → Pilot (simulate) → Validate → Expand (enforce)** model; DLP, labels, retention, and insider-risk changes are high-impact.

• **How to verify:** Confirm a named pilot scope, that simulation/test mode is available, and a change window/approval path.

• **How to implement if missing:** Define the pilot and agree change control before enforcing anything.

*The deployment model across every Purview solution is the same: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce)**, with explicit pass/fail gates and a documented rollback. Automation is delivered as **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets. Regulatory/records/legal determinations are deferred to the relevant governance functions.*

Implementation

This guide configures the archiving strategy in exact, ordered steps. **Every high-impact change is piloted on a named test scope first**, with pass/fail gates and a rollback for each step. Cmdlet names and portal paths follow Microsoft's public documentation; where a capability is plan-gated or the portal is evolving, **verify your licensing/tenant**. All write operations in the provided scripts are gated behind an explicit `-Commit` switch (default is read-only / `-WhatIf`).

Named objects used throughout (adjust to your naming standard):

Object	Name used in this guide
Pilot mailbox group (mail-enabled security group)	`grp-archive-pilot`
MRM move-to-archive tag (Default Policy Tag, all content)	`MTA-2yr` (Move to Archive, 730 days)
MRM optional personal tag	`MTA-Personal-1yr` (Move to Archive, 365 days)
MRM retention policy	`RP-Corp-Archive`
Purview DLM retention policy (keep only in pilot)	`DLM-Exchange-Keep`
Placeholder domain	`contoso.com`

> *Retention *durations* (730/365 days above) are **illustrative**. Use the values **records/compliance approved** (Requirement R5). Do **not** enable any *delete* action until records/legal has signed off.*

Step 0 — Preflight (read-only)

1. Connect and confirm access:

```
Connect-ExchangeOnline
Connect-IPPSSession
Connect-MgGraph -Scopes "User.Read.All","Group.Read.All"
```

1. Run the scope/health helper: `.\Get-ArchiveScopeHealth.ps1 -PilotGroup grp-archive-pilot`.
2. Run discovery: `.\Invoke-ArchiveDiscovery.ps1 -OutputPath .\discovery`.

Pass/fail gate: All three connections succeed; discovery reports current archive status, mailbox sizes, existing MRM tags/policies, existing DLM retention policies, and the org-level `AutoExpandingArchive` value. **If any connection fails, stop** and resolve licensing/roles (R1–R3).

Rollback: None — read-only.

Step 1 — Confirm licensing and the org-level auto-expanding archive posture

1. Verify Plan 2 / archiving entitlement on the pilot users (R1).
2. Check the org setting **without changing it yet**:

```
Get-OrganizationConfig | Format-List AutoExpandingArchive
```

1. **Decide the tiering design** with stakeholders: which mailboxes get archives, the age threshold

for move-to-archive, and whether **auto-expanding archive** is needed (only for users whose archive will exceed the base archive quota).

Pass/fail gate: Entitlement confirmed on all pilot users; the design (age threshold, scope, auto-expanding yes/no) is documented and approved.

Rollback: None — no change made.

> ***Caution:*** `Set-OrganizationConfig -AutoExpandingArchive $true` *is* ***org-wide and not reversible.*** *Prefer* ***per-mailbox*** `Enable-Mailbox -AutoExpandingArchive` *during pilot so the blast radius is limited to test users.*

Step 2 — Enable archive mailboxes for the PILOT scope only

1. Preview with the configuration script (default `-WhatIf`):

```
.\Set-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -EnableArchive -WhatIf
```

1. Apply to the pilot scope:

```
.\Set-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -EnableArchive -Commit
```

(Equivalent manual cmdlet: `Enable-Mailbox -Identity <user> -Archive`.)

1. If the design calls for auto-expanding archive on specific pilot users:

```
.\Set-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -EnableAutoExpanding -Commit
```

(Equivalent: `Enable-Mailbox -Identity <user> -AutoExpandingArchive`.)

Pass/fail gate: `Get-Mailbox <pilotuser> | fl ArchiveStatus,ArchiveGuid` shows `ArchiveStatus = Active` and a non-empty `ArchiveGuid` for every pilot user.

Rollback: `Disable-Mailbox -Identity <user> -Archive` disconnects the archive; the disconnected archive is retained for **30 days** and can be reconnected within that window, after which it is **permanently removed**. Auto-expanding archive, once enabled, **cannot be disabled**.

Step 3 — Create the MRM move-to-archive tag and retention policy

1. Preview, then create the tag and policy (script gates writes behind `-Commit`):

```
.\Set-ArchiveStrategy.ps1 -CreateMrmPolicy -TagName MTA-2yr -AgeDays 730 `
    -PolicyName RP-Corp-Archive -WhatIf
.\Set-ArchiveStrategy.ps1 -CreateMrmPolicy -TagName MTA-2yr -AgeDays 730 `
    -PolicyName RP-Corp-Archive -Commit
```

Equivalent manual cmdlets:

```
New-RetentionPolicyTag -Name "MTA-2yr" -Type All -RetentionAction MoveToArchive `
    -AgeLimitForRetention 730 -RetentionEnabled $true
New-RetentionPolicy   -Name "RP-Corp-Archive" -RetentionPolicyTagLinks "MTA-2yr"
```

1. (Optional) add a **personal** move-to-archive tag users can apply to folders:

```
New-RetentionPolicyTag -Name "MTA-Personal-1yr" -Type Personal `
    -RetentionAction MoveToArchive -AgeLimitForRetention 365 -RetentionEnabled $true
Set-RetentionPolicy -Identity "RP-Corp-Archive" `
    -RetentionPolicyTagLinks "MTA-2yr","MTA-Personal-1yr"
```

Pass/fail gate: `Get-RetentionPolicyTag "MTA-2yr"` shows `RetentionAction = MoveToArchive` and the expected `AgeLimitForRetention`; `Get-RetentionPolicy "RP-Corp-Archive"` lists the tag(s).

Rollback: `Set-RetentionPolicy -Identity RP-Corp-Archive -RetentionPolicyTagLinks @( )` unlinks the tags (stops future moves); `Remove-RetentionPolicyTag` / `Remove-RetentionPolicy` removes them. **Items already moved to the archive are not moved back.**

Step 4 — Assign the policy to the pilot mailboxes and run the Managed Folder Assistant

1. Assign and process (preview then commit):

```
.\Set-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -AssignPolicy RP-Corp-Archive -WhatIf
.\Set-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -AssignPolicy RP-Corp-Archive -Commit
.\Set-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -RunAssistant -Commit
```

Equivalent manual cmdlets:

```
Set-Mailbox -Identity <pilotuser> -RetentionPolicy "RP-Corp-Archive"
Start-ManagedFolderAssistant -Identity <pilotuser>
```

Pass/fail gate: `Get-Mailbox <pilotuser> | fl RetentionPolicy` shows `RP-Corp-Archive`; after the MFA cycle, `Get-MailboxStatistics <pilotuser> -Archive` shows a growing archive `TotalItemSize`/`ItemCount` for items older than the age threshold.

Rollback: `Set-Mailbox -Identity <user> -RetentionPolicy "Default MRM Policy"` reverts to the built-in policy; future moves stop. Already-archived items remain in the archive.

Step 5 — (Optional, keep-only in pilot) stage a Purview DLM retention policy

> *Only if records/compliance has approved a **keep** timeline. **Do not** enable *delete* in pilot.*

1. In the **Purview portal** → **Data Lifecycle Management** → **Retention policies**, or via PowerShell:

```
New-RetentionCompliancePolicy -Name "DLM-Exchange-Keep" -ExchangeLocation <pilot mailbox(es)>
New-RetentionComplianceRule   -Name "DLM-Exchange-Keep-Rule" -Policy "DLM-Exchange-Keep" `
    -RetentionDuration 2555 -RetentionComplianceAction Keep
```

(`Keep` only — no `Delete`/`KeepAndDelete` until R5 sign-off. `2555` days ≈ 7 years, illustrative.)

1. Scope the policy to the **pilot mailboxes only** for the first ring.

Pass/fail gate: `Get-RetentionCompliancePolicy "DLM-Exchange-Keep" | fl DistributionStatus` reaches a healthy/`Success` distribution state and applies to the pilot scope only.

Rollback: `Set-RetentionCompliancePolicy -Identity "DLM-Exchange-Keep" -Enabled $false` or `Remove-RetentionCompliancePolicy`. Because the action is **Keep**, no data is deleted, so rollback is safe.

Step 6 — Validate the pilot

Run the validation script and capture evidence:

```
.\Test-ArchiveStrategy.ps1 -PilotGroup grp-archive-pilot -PolicyName RP-Corp-Archive `
    -OutputPath .\validation
```

Pass/fail gate: All pilot users show archive enabled, `RP-Corp-Archive` assigned, MFA processed, archive item count > 0 where aged content exists, and (if staged) the DLM policy in a healthy state. See **Testing & Validation** for the full gate list.

Rollback: See each step above; disable/unlink rather than delete.

Step 7 — Expand (enforce) to production under change control

Only after the pilot passes and is signed off:

1. Widen the archive-enable + policy-assignment scope from `grp-archive-pilot` to the approved production population (in rings, not all at once).
1. If auto-expanding archive is org-wide by design, enable it org-wide **only now** and **only with explicit approval — remember it is** irreversible**:

```
Set-OrganizationConfig -AutoExpandingArchive $true
```

1. (Optional, tiering) enable **Microsoft 365 Archive** for inactive SharePoint sites from the **SharePoint admin center / Microsoft 365 Archive** — pay-as-you-go via an Azure subscription; **verify your licensing/tenant** as this capability is billing-gated and evolving.
1. Re-run `Test-ArchiveStrategy.ps1` and `Export-ArchiveReport.ps1` for each ring and archive the evidence.

Pass/fail gate: Each ring validates green before the next ring starts; reports show archive adoption and primary-store reduction trending as designed.

Rollback: Pause expansion; for a ring, unlink/reassign policies and disable archives per Steps 2–4 (bearing the reversibility caveats in mind). Never enable a delete action to "clean up" a rollback.

Testing & Validation

Validation proves the archiving strategy is configured, took effect, and is defensible — with **explicit pass/fail gates** and captured evidence. Run these against the **pilot scope first**, then against each expansion ring. Most checks are automated by `Test-ArchiveStrategy.ps1` (read-only); the manual cmdlets are given so results can be reproduced by hand.

> *All checks are **read-only**. None of them move or delete data.*

Gate 1 — Licensing and connectivity

Check	Method	Pass	
Admin can reach both surfaces	`Connect-ExchangeOnline` + `Connect-IPPSSession` succeed	No auth error	
Pilot users are archiving-entitled	`Get-Mailbox <user> | fl PersistedCapabilities` / license check in admin center	Plan 2 / archiving add-on present	

Fail action: resolve licensing (R1) / roles (R2) before continuing.

Gate 2 — Archive mailbox provisioned

Check	Method	Pass		
Archive enabled for every pilot user	`Get-Mailbox <user> \	fl ArchiveStatus,ArchiveGuid`	`ArchiveStatus = Active`, non-empty `ArchiveGuid`	
Auto-expanding archive matches design	`Get-Mailbox <user> \	fl AutoExpandingArchiveEnabled and Get-OrganizationConfig \	fl AutoExpandingArchive`	Matches the approved design

Gate 3 — MRM tag and policy correct

Check	Method	Pass	
Move-to-archive tag exists and is correct	`Get-RetentionPolicyTag "MTA-2yr" \	fl RetentionAction,AgeLimitForRetention,RetentionEnabled`	`RetentionAction = MoveToArchive`, expected age, `RetentionEnabled = True`
Policy links the tag	`Get-RetentionPolicy "RP-Corp-Archive" \	fl RetentionPolicyTagLinks`	Includes `MTA-2yr`
Policy assigned to pilot mailboxes	`Get-Mailbox <user> \	fl RetentionPolicy`	`RP-Corp-Archive`

Gate 4 — Managed Folder Assistant processed and items moved

Check	Method	Pass	
MFA has processed the mailbox	Confirm the MFA work cycle completed (or `Start-ManagedFolderAssistant` was run) and no error	Processed without error	
Aged items are in the archive	`Get-MailboxStatistics <user> -Archive \	fl ItemCount,TotalItemSize` before vs. after	Archive `ItemCount`/`TotalItemSize` increased for users with content older than the age threshold
Primary store reduced	Compare `Get-MailboxStatistics <user>` primary size before/after	Primary `TotalItemSize` trending down where content aged out	

If aged content exists but nothing moved, MFA may not have run yet (it is throttled/asynchronous), or the tag age/enabled flag is wrong — see Troubleshooting.

Gate 5 — Purview DLM retention policy (if staged)

Check	Method	Pass	
Policy distributed successfully	`Get-RetentionCompliancePolicy "DLM-Exchange-Keep" | fl DistributionStatus,Enabled`	Healthy/`Success`, `Enabled = True`
Action is Keep only in pilot	`Get-RetentionComplianceRule -Policy "DLM-Exchange-Keep" | fl RetentionComplianceAction`	`Keep` (no delete in pilot)
Scope is pilot-only	`Get-RetentionCompliancePolicy "DLM-Exchange-Keep" | fl ExchangeLocation`	Pilot mailboxes only

Gate 6 — No unintended deletion / hold conflict

Check	Method	Pass
No delete action active in pilot	Review all MRM tags in `RP-Corp-Archive` and the DLM rule	No `DeleteAndAllowRecovery` / `Delete` / `KeepAndDelete` enabled
Legal holds respected	Confirm with legal that any holds are in place and unaffected	Holds intact; archiving does not bypass hold

Fail action for any gate: stop expansion, record the failing gate, apply the matching Troubleshooting fix, re-run the gate, and only proceed when green.

Evidence to capture

- Discovery output (baseline) and validation output (post-change) from the scripts.
- `Get-Mailbox`/`Get-MailboxStatistics` before/after for a sample of pilot users (archive growth, primary reduction).
- The MRM tag/policy definitions and the DLM policy distribution status.
- The adoption/tiering report from `Export-ArchiveReport.ps1`.
- Records/compliance/legal sign-off reference for the retention timeline (R5).

Store the evidence with the change record so the archiving strategy is **defensible** on audit.

Troubleshooting

Common failure signatures when standing up the archiving strategy, and how to resolve them. All diagnostic cmdlets below are **read-only**.

Archive mailbox will not enable

Signature: `Enable-Mailbox -Archive` errors, or `ArchiveStatus` stays `None`.

- **Licensing.** Archiving requires **Exchange Online Plan 2** (E3/E5) or **Plan 1 + Exchange Online Archiving add-on** — *verify your licensing/tenant*. Confirm the user's assigned plan in the admin center, then retry. (R1)
- **Role.** You need Recipient Management (mail-recipient) rights. Check `Get-ManagementRoleAssignment -RoleAssignee <you>`. (R2)
- **Replication lag.** After enabling, `ArchiveStatus` and `ArchiveGuid` can take time to populate. Re-check `Get-Mailbox <user> | fl ArchiveStatus,ArchiveGuid` after a short wait.

Auto-expanding archive can't be turned off

Signature: you enabled `-AutoExpandingArchive` and now need to reverse it.

- **It is not reversible.** Auto-expanding archive **cannot be disabled** once enabled, per Microsoft.

This is why the Implementation Guide enables it per-mailbox in pilot and only org-wide at expansion with explicit approval. There is no fix — plan around it.

Items are not moving to the archive

Signature: archive is enabled and the policy is assigned, but `Get-MailboxStatistics -Archive` shows `ItemCount = 0` or no growth.

- **The Managed Folder Assistant hasn't run.** MFA is asynchronous and throttled. Force it on a pilot mailbox: `Start-ManagedFolderAssistant -Identity <user>`, then re-check after it completes.
- **Tag age not reached.** `MoveToArchive` only moves items **older than** `AgeLimitForRetention`. If the mailbox has no content older than the threshold, nothing moves — that is correct. Verify with a user who has aged mail, or temporarily test with a smaller age on a **test** mailbox.
- **Tag disabled or wrong action.** Confirm `Get-RetentionPolicyTag <tag> | fl RetentionAction,RetentionEnabled` shows `MoveToArchive` and `True`.
- **Tag not linked / policy not assigned.** Confirm `Get-RetentionPolicy <policy>` lists the tag and `Get-Mailbox <user> | fl RetentionPolicy` shows the policy.
- **Item on hold.** Litigation/In-Place hold or Purview retention can keep an item in place; this is expected and takes precedence over the archive move.

Retention policy assigned but behaviour unchanged

Signature: `Set-Mailbox -RetentionPolicy` succeeded but nothing happens.

- Retention-policy changes only take effect on the **next MFA cycle**. Trigger MFA and wait.
- Confirm the assignment actually stuck: `Get-Mailbox <user> | fl RetentionPolicy`.

Purview DLM retention policy stuck or not applying

Signature: `Get-RetentionCompliancePolicy | fl DistributionStatus` shows a pending/error state.

- **Distribution takes time.** New DLM policies can take time to reach all locations; recheck later.
- **Connect the right session.** DLM cmdlets require **Security & Compliance PowerShell** (`Connect-IPPSSession`), *not* the Exchange Online session.
- **Scope/location.** Confirm `ExchangeLocation` includes the intended (pilot) mailboxes and nothing unintended.
- **Never add a delete action to "fix" it.** Keep the action `Keep` until records/legal signs off (R5). If a delete action appears in pilot, it was misconfigured — remove it.

`Get-RetentionCompliancePolicy` / DLM cmdlets not found

Signature: command not recognized.

- You are in the wrong session. Run `Connect-IPPSSession` (Security & Compliance PowerShell). Use `Get-ArchiveScopeHealth.ps1` to confirm both sessions are connected.

Pilot group resolves to zero mailboxes

Signature: scripts report `0 mailbox member(s)`.

- Confirm the group name/identity and that its members are **mailboxes** (not contacts or nested groups the cmdlet doesn't expand). Check `Get-DistributionGroupMember -Identity <group>`.

Script made no change

Signature: `Set-ArchiveStrategy.ps1` printed a plan but changed nothing.

- **Working as designed.** The script is `-WhatIf` by default. Re-run with `-Commit` to apply, after reviewing the preview.

If a **Testing & Validation** gate keeps failing after these steps, stop the expansion, capture the discovery/validation output, and review against the Implementation Guide's pass/fail gate for that step before proceeding. Remember the reversibility limits — never enable a delete action as a "cleanup" during troubleshooting.

Best Practices

Practices that keep the archiving strategy healthy, cost-effective, and defensible over time. They extend the category Global Requirements and the pilot-first delivery model.

Strategy and scope

- **Tier by access frequency, not by guesswork.** Base the move-to-archive age on real telemetry (mailbox growth, how far back users actually search) from the Discovery report, not a round number.
- **Separate "move" from "delete" completely.** Archiving (move-to-archive) reduces primary-store cost and is low-risk; **disposition (delete) is a records/legal decision** and irreversible. Ship archiving first; only add deletion once records/compliance has approved a schedule (R5).
- **Right-size auto-expanding archive.** Enable auto-expanding archive **only** for users whose archive will exceed the base quota — it is **irreversible** and adds background processing. Prefer per-mailbox enablement over the org-wide switch during rollout.

Change discipline

- **Always pilot on test mailboxes first.** Never make a production user the first ring. Archiving moves data and is hard to reverse; a bad age threshold on production is expensive to unwind.
- **Gate every write behind `-Commit`.** Run every script in `-WhatIf` first, read the plan, then commit. Keep the default read-only.
- **One change per ring, validated before the next.** Enable archives, assign the policy, run MFA, validate — then widen scope. Do not stack changes across rings.
- **Document the rollback before you change anything.** For each step know the reverse action and its limits (disconnected archives are kept 30 days; auto-expanding cannot be disabled; already-moved items do not come back).

Operations and monitoring

- **Understand the Managed Folder Assistant cadence.** MFA is asynchronous and throttled; archiving is not instant. Use `Start-ManagedFolderAssistant` to force a pilot mailbox, but expect the org-wide work cycle to process at its own pace.
- **Report on adoption and reduction, not just configuration.** Track archive item growth and primary-store reduction over time (`Export-ArchiveReport.ps1`) — that is the value evidence and the early-warning signal if MFA stalls.
- **Watch for hold interactions.** Litigation hold, In-Place Hold, and Purview retention can keep content beyond the archive move; that is correct behavior. Never disable a hold to make archiving "look cleaner."
- **Re-baseline periodically.** Re-run Discovery quarterly; mailbox growth and usage patterns drift, and the age threshold may need tuning.

Governance and defensibility

• **Keep records/compliance/legal in the loop.** The retention timeline and any disposition are their determinations; capture named sign-off and store it with the change evidence.

• **Least privilege, no standing Global Admin.** Use Recipient Management / Retention Management scoped roles; reserve Global Administrator for the specific org-wide actions that require it.

• **No secrets in automation.** The scripts use interactive modern-auth (`Connect-ExchangeOnline` / `Connect-IPPSSession` / `Connect-MgGraph`) — never embed credentials, tokens, or app secrets.

• **Treat plan/portal detail as verify-first.** Archiving entitlements, auto-expanding archive, and **Microsoft 365 Archive** tiering are licensing-/billing-gated and evolving — confirm availability and paths against current Microsoft documentation for your tenant before relying on them.

• **Retain the evidence trail.** Baseline + post-change statistics, policy definitions, MFA/validation output, and sign-off together make the archiving strategy auditable and defensible.

Deliverables Summary

Everything included in this solution and how the pieces fit together.

Documentation

Deliverable	Purpose
Solution Overview	What the archiving strategy delivers, how mailbox archive / retention / tiering work in Purview and Exchange Online, and the Assess → Design → Configure → Pilot → Validate → Expand model.
Requirements Guide	Solution-specific prerequisites (archiving licensing, least-privilege roles, MRM/MFA toolchain, in-scope mailboxes and tiering intent, records/legal sign-off, pilot scope) referencing the category Global Requirements.
Implementation Guide	Exact ordered steps to enable archive mailboxes, create the `MTA-2yr` move-to-archive tag and `RP-Corp-Archive` policy, assign to the pilot, run the Managed Folder Assistant, optionally stage the `DLM-Exchange-Keep` retention policy, validate, and expand — each step with pass/fail gates and rollback.
Testing & Validation	Six pass/fail gates proving the archiving strategy took effect, with the evidence to capture.
Troubleshooting	Common failure signatures (archive not provisioning, items not moving, MFA lag, policy not distributing) and resolutions.

Deliverable	Purpose
Best Practices	Keeping the strategy healthy, cost-effective, and defensible; move-vs-delete separation; change discipline.
Executive Summary	One-page business framing for decision-makers.

PowerShell scripts (5)

All scripts use interactive modern-auth (`Connect-ExchangeOnline` / `Connect-IPPSSession` / `Connect-MgGraph`), contain **no credentials or secrets**, and are **read-only or `-WhatIf` by default** — writes require an explicit `-Commit` switch.

Script	Role	Default safety
`Get-ArchiveScopeHealth.ps1`	Scope/health helper — connectivity, module, role and pilot-group preflight	Read-only
`Invoke-ArchiveDiscovery.ps1`	Discovery — archive status, mailbox sizes, MRM tags/policies, DLM retention policies, org auto-expanding setting	Read-only
`Set-ArchiveStrategy.ps1`	Configuration/Remediation — enable archives, create/assign MRM tag & policy, run MFA	`-WhatIf` unless `-Commit`
`Test-ArchiveStrategy.ps1`	Validation — runs the pass/fail gates and reports PASS/FAIL	Read-only
`Export-ArchiveReport.ps1`	Reporting — adoption, archive growth, and primary-store reduction to CSV/HTML	Read-only

How to use them together

1. `Get-ArchiveScopeHealth.ps1` — confirm you can proceed (licensing, roles, modules, pilot group).
2. `Invoke-ArchiveDiscovery.ps1` — capture the **baseline**.
3. `Set-ArchiveStrategy.ps1 -WhatIf` → review → `-Commit` — configure the **pilot** scope.
4. `Test-ArchiveStrategy.ps1` — confirm the pilot passes every gate.
5. `Export-ArchiveReport.ps1` — evidence of adoption and reduction; repeat per expansion ring.

Boundaries

• The **retention timeline and any disposition (delete)** are **records / compliance / legal

determinations** — deferred to those functions; this solution provides the technical method and evidence only.

• Purview/Exchange **licensing, auto-expanding archive, and Microsoft 365 Archive tiering** are plan-/billing-gated and evolving — **verify your licensing/tenant**.

• Automation is **PowerShell** (Exchange Online + Security & Compliance PowerShell / Graph); no hardcoded credentials.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

Export-ArchiveReport.ps1

Get-ArchiveScopeHealth.ps1

Invoke-ArchiveDiscovery.ps1

Set-ArchiveStrategy.ps1

Test-ArchiveStrategy.ps1